Independent native replay of the optimization outputs

2026-10-12. Correct exact model: `icv1-f2m83-t6151469093347-9eba6cd5`, field low terms `[0,2,4,7]`, curve $y^2+xy=x^3+x^2+1$, subgroup prime `8569786107849059`. This report describes arithmetic and file-provenance verification of completed outputs. It adds no timing measurement or algebraic discovery.

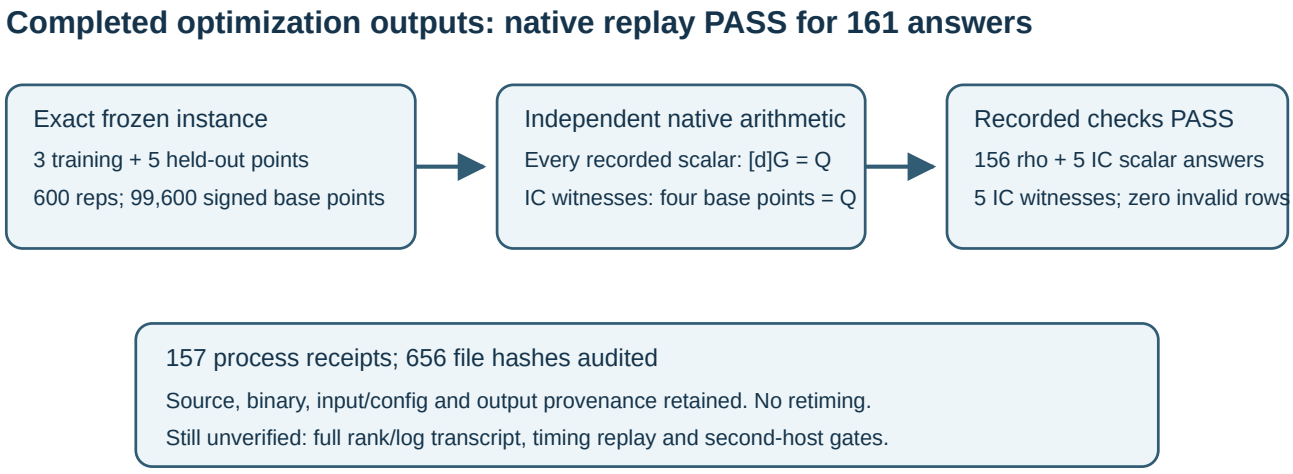


Figure 1: Completed answers were replayed after all timed children finished. Repeated training and rho controls are counts of answers, not independent-target success-rate samples.

Recorded arithmetic result

The parent ran the prepared standalone C++ checker after timings completed. `final_receipt.json` reports **PASS**: **156 rho scalar answers, five IC scalar answers and five IC four-point witnesses** checked independently, with zero invalid rows.

Output group	Distinct frozen points	Scalar-answer replays	Four-point witnesses
Rho training/kernel/policy	3	126/126	Not applicable
Rho holdout/default/selected	5	30/30	Not applicable
IC holdout	The same 5	5/5	5/5

All 157 process dispositions are preserved as `COMPLETED_VERIFIED`. There were 156 rho children and one IC child with five target rows. Each of the three training points appears in 42 rho records; each holdout point appears in six rho records plus its one IC record. The eight-point freeze and generation law are recorded in the preserved `GENERATION_REPORT.md/PDF`. These counts describe retained output checks; they do not certify an IID target population, a timing ratio or a statistical algorithm winner.

The final checker reproduced the eight-fixture SplitMix/rejection stream, subgroup membership, `[d]G=Q` fixture certificates, disjoint signed-orbit exclusions, individual point-file hashes and five-row holdout file. The fixture-manifest and validation-sidecar hashes match the earlier freeze. No point, scalar sidecar, retained base or old receipt was rewritten.

What each answer check proves

The native implementation validates the exact retained field, G, subgroup and 99,600-point factor base with its 600 signed Frobenius orbit columns. For every recovered rho or IC scalar it recomputes `[d]G` with standalone polynomial-field affine arithmetic and compares the result to that row's exact frozen public point. The scalar sidecar is compared only after the answer's point replay. Measured-child inputs remain point-only.

For all five IC answers it checks the four actual point indices against the retained base, compares their x codes and both intermediate sum abscissae, and independently verifies `P1+P2+P3+P4=Q`. Orbit labels and signs were checked against the full base before using those indices. Caps, failures or missing answers would remain explicit records rather than certified answers; none were present in these completed outputs.

The earlier `cdcc5432` historical model alias describes another modulus. The exact-field checker and these outputs use `9eba6cd5` and `[0,2,4,7]`. The parent study retains the historical identity erratum. No new identity or library-arithmetic result is asserted here: these are applications of the existing independent arithmetic to the recorded answers.

File provenance and remaining limits

The post-run native SHA-256 audit also passed. It audited 157 execution receipts and hashed 157 producer files, retaining 656 distinct file hashes. It compared every recorded executable, existing argument input/config, semantic output, stdout and stderr hash with actual bytes. Two executable SHA-256 identities covered the rho and IC children. All 157 recorded known-answer environment flags were false. The audit is preserved as `final_file_hash_audit.json` with source `verify_file_provenance.cpp`.

This hash audit establishes consistency of the retained file provenance; it does not independently observe historical process behavior or certify every possible branch of library field arithmetic. The final answer replay validates the recorded target outputs, not a universal arithmetic identity.

The full 600-row rank/factor-log transcript remains unverified. The retained IC target and summary outputs do not contain every setup query, admitted relation witness, echelon matrix and solved factor-log vector. Independent reproduction of operation counts, timings, host isolation and second-host promotion gates also remain separate. The parent evaluation owns the statistical report and interpretation; this report makes no speedup claim.

Evidence and reproduction

Final arithmetic receipt SHA-256:

`6640a9f69e8ef66f3e31a3eab70c18925bdc8140a623aac3559c01fdeb57e294`

Native verifier sources are `check_optimization.cpp`, `native_helpers.hpp` and `check.cpp`. Source/receipt and executable hashes are preserved beside the report; binaries are excluded from the repository. Use the compile and replay instructions in `README.md` and create a new output receipt path. Run verifiers only after timed children finish. `final_receipt.json` and all timed records were read without modification during this review.

The editable diagram is `fresh_verification.svg`, with its vector PDF rendering. The original fixture-generation report and hash manifest are preserved; the final manifest adds completed receipts and documentation.